Statistical Inference on Global Health Data: A comprehensive analysis of the WHO Life Expectancy Dataset

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The dataset was loaded successfully. We proceeded to do some checks onto the dataset.

```r
# Checking for dimensions
cat("Dimensions:", dim(df), "\n\n")
```

```
## Dimensions: 2864 21
```

```r
# Checking column names
names(df)
```

```
##  [1] "Country"                      "Region"
##  [3] "Year"                         "Infant_deaths"
##  [5] "Under_five_deaths"            "Adult_mortality"
##  [7] "Alcohol_consumption"          "Hepatitis_B"
##  [9] "Measles"                      "BMI"
## [11] "Polio"                        "Diphtheria"
## [13] "Incidents_HIV"                "GDP_per_capita"
## [15] "Population_mln"               "Thinness_ten_nineteen_years"
## [17] "Thinness_five_nine_years"     "Schooling"
## [19] "Economy_status_Developed"     "Economy_status_Developing"
## [21] "Life_expectancy"
```

```r
# Checking data types
str(df)
```

```
## 'data.frame':    2864 obs. of  21 variables:
##  $ Country                    : chr  "Turkiye" "Spain" "India" "Guyana" ...
##  $ Region                     : chr  "Middle East" "European Union" "Asia" "South America" ...
##  $ Year                       : int  2015 2015 2007 2006 2012 2006 2015 2000 2001 2008 ...
##  $ Infant_deaths              : num  11.1 2.7 51.5 32.8 3.4 9.8 6.6 8.7 22 15.3 ...
##  $ Under_five_deaths          : num  13 3.3 67.9 40.5 4.3 11.2 8.2 10.1 26.1 17.8 ...
##  $ Adult_mortality            : num  105.8 57.9 201.1 222.2 58 ...
##  $ Alcohol_consumption        : num  1.32 10.35 1.57 5.68 2.89 ...
##  $ Hepatitis_B                : int  97 97 60 93 97 88 97 88 97 97 ...
##  $ Measles                    : int  65 94 35 74 89 86 97 99 87 92 ...
##  $ BMI                        : num  27.8 26 21.2 25.3 27 26.4 26.2 25.9 27.9 26.5 ...
##  $ Polio                      : int  97 97 67 92 94 89 97 99 97 96 ...
##  $ Diphtheria                 : int  97 97 64 93 94 89 97 99 99 90 ...
##  $ Incidents_HIV              : num  0.08 0.09 0.13 0.79 0.08 0.16 0.08 0.08 0.13 0.43 ...
##  $ GDP_per_capita             : int  11006 25742 1076 4146 33995 9110 9313 8971 3708 2235 ...
##  $ Population_mln             : num  78.53 46.44 1183.21 0.75 7.91 ...
##  $ Thinness_ten_nineteen_years: num  4.9 0.6 27.1 5.7 1.2 2 2.3 2.3 4 2.9 ...
##  $ Thinness_five_nine_years   : num  4.8 0.5 28 5.5 1.1 1.9 2.3 2.3 3.9 3.1 ...
##  $ Schooling                  : num  7.8 9.7 5 7.9 12.8 7.9 12 10.2 9.6 10.9 ...
##  $ Economy_status_Developed   : int  0 1 0 0 1 0 0 1 0 0 ...
##  $ Economy_status_Developing  : int  1 0 1 1 0 1 1 0 1 1 ...
##  $ Life_expectancy            : num  76.5 82.8 65.4 67 81.7 78.2 71.2 71.2 71.9 68.7 ...
```

```r
# Checking for null values
colSums(is.na(df))
```

```
##                     Country                      Region
##                           0                           0
##                        Year               Infant_deaths
##                           0                           0
##           Under_five_deaths             Adult_mortality
##                           0                           0
##         Alcohol_consumption                 Hepatitis_B
```

```
##                            0                             0
##                      Measles                           BMI
##                            0                             0
##                        Polio                    Diphtheria
##                            0                             0
##                Incidents_HIV                 GDP_per_capita
##                            0                             0
##               Population_mln Thinness_ten_nineteen_years
##                            0                             0
##     Thinness_five_nine_years                     Schooling
##                            0                             0
##     Economy_status_Developed   Economy_status_Developing
##                            0                             0
##              Life_expectancy
##                            0
```

We saw that there 2864 rows and 21 columns in the dataset.
There were no missing values in all the variables. In order to be sure, we also proceeded to check if there were rows labeled 'null'.

```r
# Checking if there were values masked as 'null', 'NULL' or " "
cat("Searching for text based masked nulls \n")
```

```
## Searching for text based masked nulls
```

```r
colSums(df == "null" | df == "NULL" | df == "", na.rm = TRUE)
```

```
##                      Country                        Region
##                            0                             0
##                         Year                 Infant_deaths
##                            0                             0
##             Under_five_deaths               Adult_mortality
##                            0                             0
##          Alcohol_consumption                   Hepatitis_B
##                            0                             0
##                      Measles                           BMI
##                            0                             0
##                        Polio                    Diphtheria
##                            0                             0
##                Incidents_HIV                 GDP_per_capita
##                            0                             0
##               Population_mln Thinness_ten_nineteen_years
##                            0                             0
##     Thinness_five_nine_years                     Schooling
##                            0                             0
##     Economy_status_Developed   Economy_status_Developing
##                            0                             0
##              Life_expectancy
##                            0
```

```r
# Checking for suspicious zeros (0) in the health metrics
cat("\n Investigating null values masquerading as 0s \n")
```

```
##
##  Investigating null values masquerading as 0s
```

```r
cat("Zeros in Life Expectancy:", sum(df$Life_expectancy == 0, na.rm = TRUE), "\n" )
```

```
## Zeros in Life Expectancy: 0
```

```r
cat("Zeros in under 5 deaths:", sum(df$Under_five_deaths == 0, na.rm = TRUE), "\n")
```

```
## Zeros in under 5 deaths: 0
```

```r
cat("Zeros in under Adult Mortality:", sum(df$Adult_mortality == 0, na.rm = TRUE), "\n")
```

```
## Zeros in under Adult Mortality: 0
```

```r
cat("Zeros in under alcohol consumption:", sum(df$Alcohol_consumption == 0, na.rm = TRUE),
 ↪  "\n")
```

```
## Zeros in under alcohol consumption: 38
```

```r
cat("Zeros in under BMI:", sum(df$BMI == 0, na.rm = TRUE), "\n")
```

```
## Zeros in under BMI: 0
```

```r
cat("Zeros in under Life Expectancy:", sum(df$Life_expectancy == 0, na.rm = TRUE), "\n")
```

```
## Zeros in under Life Expectancy: 0
```

```r
cat("Zeros in under GDP per capita:", sum(df$GDP_per_capita == 0, na.rm = TRUE), "\n")
```

```
## Zeros in under GDP per capita: 0
```

```r
cat("Zeros in incidents in HIV:", sum(df$Incidents_HIV == 0, na.rm = TRUE), "\n")
```

```
## Zeros in incidents in HIV: 0
```

It was seen that there were no masked values as "null", 'null' or an empty string.
Upon investigating the zeros, there was no suspicious activity of zeros. The selected columns for this check were guided by real world scenarios. This is such as no country could have a GDP of zero or an entire population could have a BMI of zero. We did not check for the various diseases because it is possible for the various developed countries to have incidences where there are no certain diseases because of their well developed health care system which can also be backed by their robust immunization procedures. With regards to alcohol where we obtained 38 zeros, we needed to identify if these countries came from the Muslim majority countries or countries under an Islamic regime of which according to their religious views, consumption of alcohol is prohibited.

```r
# Checking which countries have an alcohol consumption of zero
unique(df$Country[df$Alcohol_consumption == 0])
```

```
## [1] "Saudi Arabia" "Kuwait"       "Somalia"      "Bangladesh"   "Mauritania"
## [6] "Afghanistan"  "Libya"
```

The Muslim majority hypothesis proved to be correct as the countries with 0 alcohol consumption came from Islamic nations.
We needed to check if the highly-developed countries indeed had no vaccine preventable diseases. We also proceeded to investigate if presence of zeros meant that no immunization coverage was done or it meant that there were no such cases.

```r
# Checking if it was immunization coverage or case count
cat("Summary of disease columns \n")
```

```
## Summary of disease columns
```

```r
summary(df[ , c("Measles", "Polio", "Diphtheria")])
```

```
##     Measles           Polio         Diphtheria
##  Min.   :10.00   Min.   : 8.0   Min.   :16.00
##  1st Qu.:64.00   1st Qu.:81.0   1st Qu.:81.00
```

```
##  Median :83.00   Median :93.0   Median :93.00
##  Mean   :77.34   Mean   :86.5   Mean   :86.27
##  3rd Qu.:93.00   3rd Qu.:97.0   3rd Qu.:97.00
##  Max.   :99.00   Max.   :99.0   Max.   :99.00
```

An inspection of these diseases proved that there was a strict maximum threshold of 99. For actual cases, this value would scale to higher figures such as tens of thousands. This evidently proved that these values were actual percentage coverage rather than actual case counts.
We needed to do an investigation of these percentages. This was to check whether the percentages meant 99% covered or not covered. This was done by the use of a correlation. If it meant covered, there would be an inverse relationship between the disease and `Infant_deaths` or `Under_five_deaths`.

```r
cat("The correlation between the Measles disease (coverage) and death of
    children under 5 years is", cor(df$Measles, df$Under_five_deaths, use = 'complete.obs'),
 ↪  "\n\n")
```

```
## The correlation between the Measles disease (coverage) and death of
##     children under 5 years is -0.5129717
```

```r
cat("The correlation between the Polio disease (coverage) and death of
    infants is", cor(df$Polio, df$Infant_deaths, use = 'complete.obs'), "\n\n")
```

```
## The correlation between the Polio disease (coverage) and death of
##     infants is -0.7407905
```

This showed an inverse relationship between the diseases and the child mortality. This meant that the values exhibited by the diseases meant the extensiveness of the vaccinations. A 99% coverage meant that most parts of this particular country was vaccinated against the specific disease.

The project's next step was to check if indeed the high immunization coverage came from the highly developed countries. This would validate our hypothesis of developed countries having better healthcare. We began by investigating the robustness of `Economy_status_Developing` and `Economy_status_Developed` features to ensure that they were mutually exclusive.

```r
# Investigating that the two economy columns do not add up to 2. Mutual
# exclusivity is needed
economy_check <- df$Economy_status_Developing + df$Economy_status_Developed

# Summary of the calculation
cat("Summary of the economy checks \n")
```

```
## Summary of the economy checks
```

```r
summary(economy_check)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       1       1       1       1       1       1
```

```r
# Tabular form of these two columns
cat("\n Cross validation table of the two economy features \n")
```

```
##
##  Cross validation table of the two economy features
```

```r
table(Developed = df$Economy_status_Developed, Developing = df$Economy_status_Developing)
```

```
##          Developing
## Developed    0    1
```

```
##          0    0 2272
##          1  592    0
```

We saw that the two columns were indeed mutually exclusive as they both did not add up to 2. The cross validation table backs this up whereby there are no individual countries where they are both developed and developing. This is given by the `0`. Additionally, the table also informs us that there are `2272` records from developing countries and there are `592` records from developed countries.

We now proceeded to see if the high vaccination coverage came from developed or developing countries.

```r
# Comparing the coverage between the two economy statuses
cat("Average coverage: Developing (0) vs Developed (1) \n")
```

```
## Average coverage: Developing (0) vs Developed (1)
```

```r
aggregate(cbind(Measles, Polio, Diphtheria) ~ Economy_status_Developed, data = df, FUN = mean,
  na.rm = TRUE)
```

```
##   Economy_status_Developed  Measles    Polio Diphtheria
## 1                        0 74.50044 84.31954   83.97711
## 2                        1 88.26182 94.86655   95.07770
```

```r
# Checking which economy status holds 0% coverage record for measles
cat("Economy status for countries with 0% Measles coverage \n")
```

```
## Economy status for countries with 0% Measles coverage
```

```r
table(Measles_zero = df$Measles == 0, Developed = df$Economy_status_Developed)
```

```
##             Developed
## Measles_zero    0    1
##        FALSE 2272  592
```

We saw that developed countries had a better coverage of vaccination as compared to developing countries. For instance, Measles had a 88.26% coverage in developed countries while it had a 74.5% coverage in the developing countries. This trend was seen in Polio and Diphtheria.
The table also showed that there was no instance of any country where it had a 0% coverage of Measles. Every single row returned `FALSE` thus proving coverage across all countries.
We then proceeded to conduct descriptive statistics.

```r
# Isolating the numeric columns as well as omitting variables which
# would be less meaningful to have summary statistics
cols_to_exclude <- c("Year", "Economy_status_Developed", "Economy_status_Developing")
num_df <- df[ , sapply(df, is.numeric)]
num_df <- num_df [, !(names(num_df) %in% cols_to_exclude)]

# Calculating the statistics
stats <- function(x) {
  x <- na.omit(x)
  c(
    Mean = mean(x),
    Median = median(x),
    Variance = var(x),
    SD = sd(x),
    Range = max(x) - min(x),
    IQR = IQR(x)
  )
}
```

```r
stats_table <- t(sapply (num_df, stats))

cat ("Descriptive statistics \n")
```

```
## Descriptive statistics
```

```r
round(stats_table, 2)
```

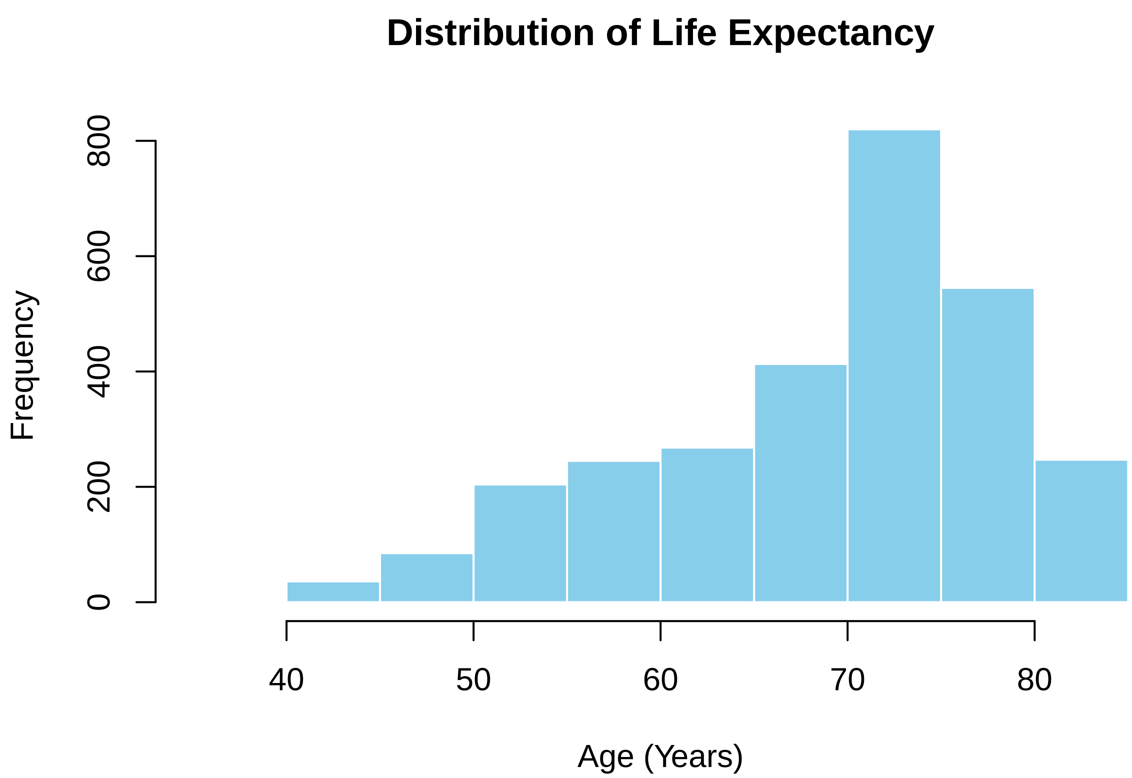
```
##                                Mean  Median     Variance       SD     Range
## Infant_deaths                 30.36   19.60       758.35    27.54    136.30
## Under_five_deaths             42.94   23.10      1986.48    44.57    222.60
## Adult_mortality              192.25  163.84     13204.37   114.91    669.98
## Alcohol_consumption            4.82    4.02        15.86     3.98     17.87
## Hepatitis_B                   84.29   89.00       255.86    16.00     87.00
## Measles                       77.34   83.00       348.18    18.66     89.00
## BMI                           25.03   25.50         4.81     2.19     12.30
## Polio                         86.50   93.00       227.42    15.08     91.00
## Diphtheria                    86.27   93.00       241.31    15.53     83.00
## Incidents_HIV                  0.89    0.15         5.67     2.38     21.67
## GDP_per_capita             11540.92 4217.00 286787076.14 16934.79 112270.00
## Population_mln                36.68    7.85     18628.39   136.49   1379.78
## Thinness_ten_nineteen_years    4.87    3.30        19.70     4.44     27.60
## Thinness_five_nine_years       4.90    3.40        20.48     4.53     28.50
## Schooling                      7.63    7.80        10.06     3.17     13.00
## Life_expectancy               68.86   71.40        88.47     9.41     44.40
##                                 IQR
## Infant_deaths                 39.25
## Under_five_deaths             56.33
## Adult_mortality              139.88
## Alcohol_consumption            6.58
## Hepatitis_B                   18.00
## Measles                       29.00
## BMI                            3.20
## Polio                         16.00
## Diphtheria                    16.00
## Incidents_HIV                  0.38
## GDP_per_capita             11141.25
## Population_mln                21.59
## Thinness_ten_nineteen_years    5.60
## Thinness_five_nine_years       5.70
## Schooling                      5.20
## Life_expectancy               12.70
```

From the summary statistics above, we saw no major outliers or anomalies that would have warranted an immediate action. Although, there was possible skewness `GDP_per_capita` whereby it had a median of *4217.00* and a mean of *11540.92*. A group of high income individuals would cause this. Such trends were also witnessed in `Population_mln` and `Incidents_HIV`, where the former could be explained by countries having a high population. With that, we proceeded to generate visualizations which would help us see missed information.

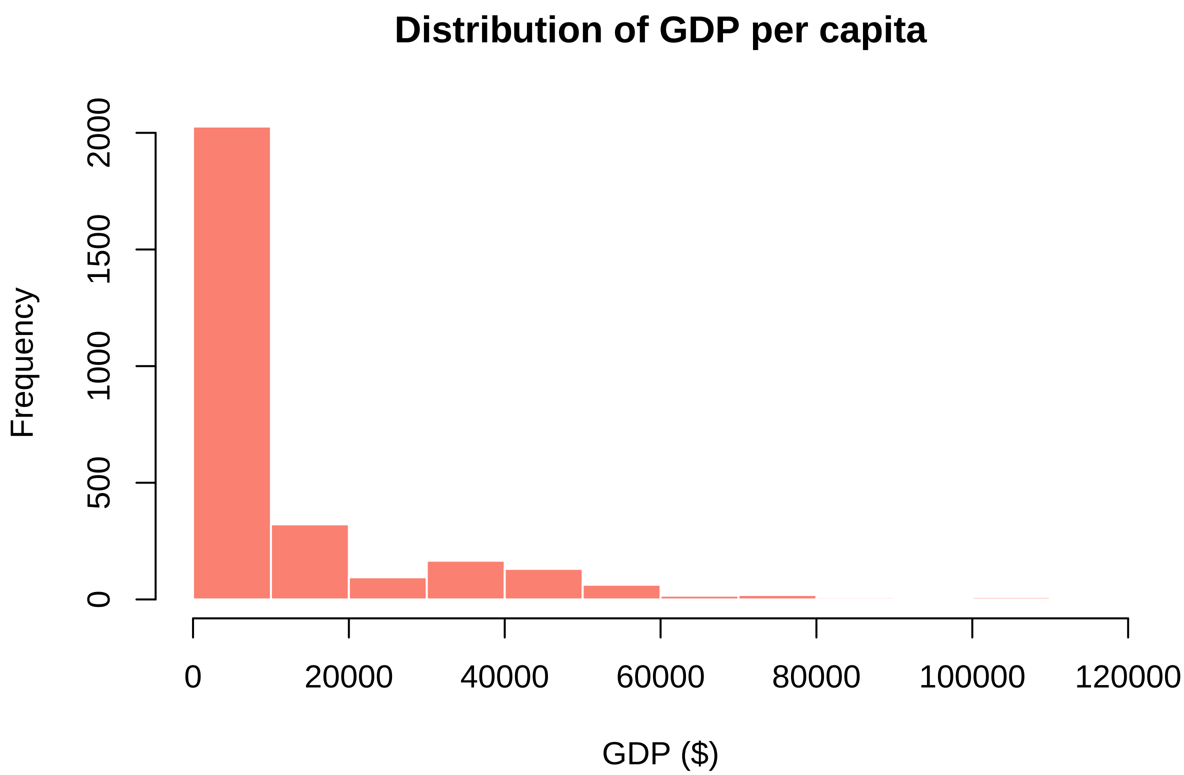
We began by histograms:

```r
# Life Expectancy
hist(df$Life_expectancy,
     main = "Distribution of Life Expectancy",
     xlab = "Age (Years)",
```

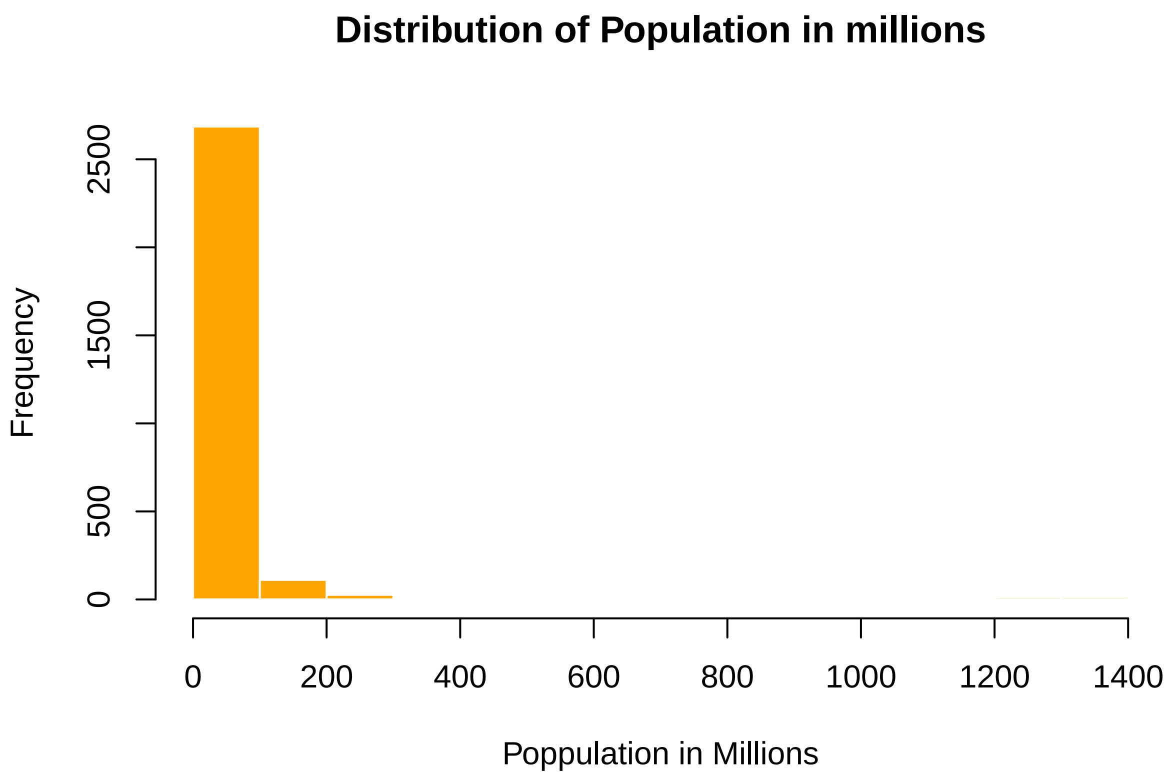
```
     col = "skyblue",
     border = "white")
```



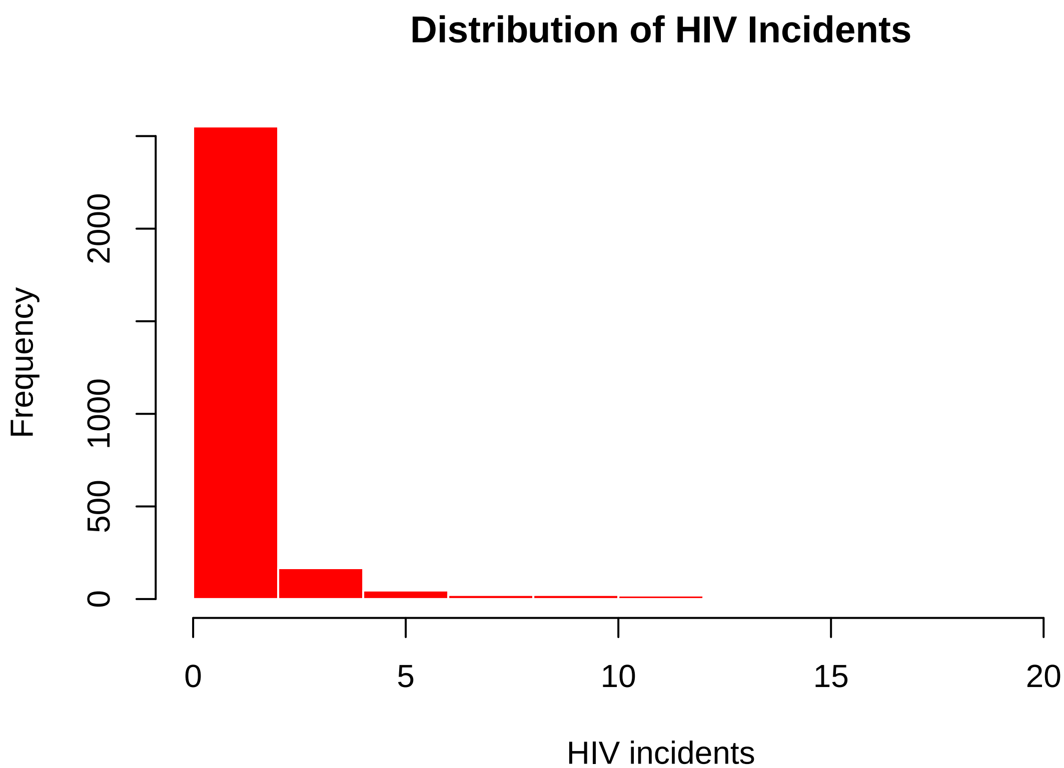

```
# GDP Per Capita
hist(df$GDP_per_capita,
     main = "Distribution of GDP per capita",
     xlab = "GDP ($)",
     col = "salmon",
     border = "white")
```




```r
# Population in Million
hist(df$Population_mln,
     main = "Distribution of Population in millions",
     xlab = "Poppulation in Millions",
     col = "orange",
     border = "white")
```



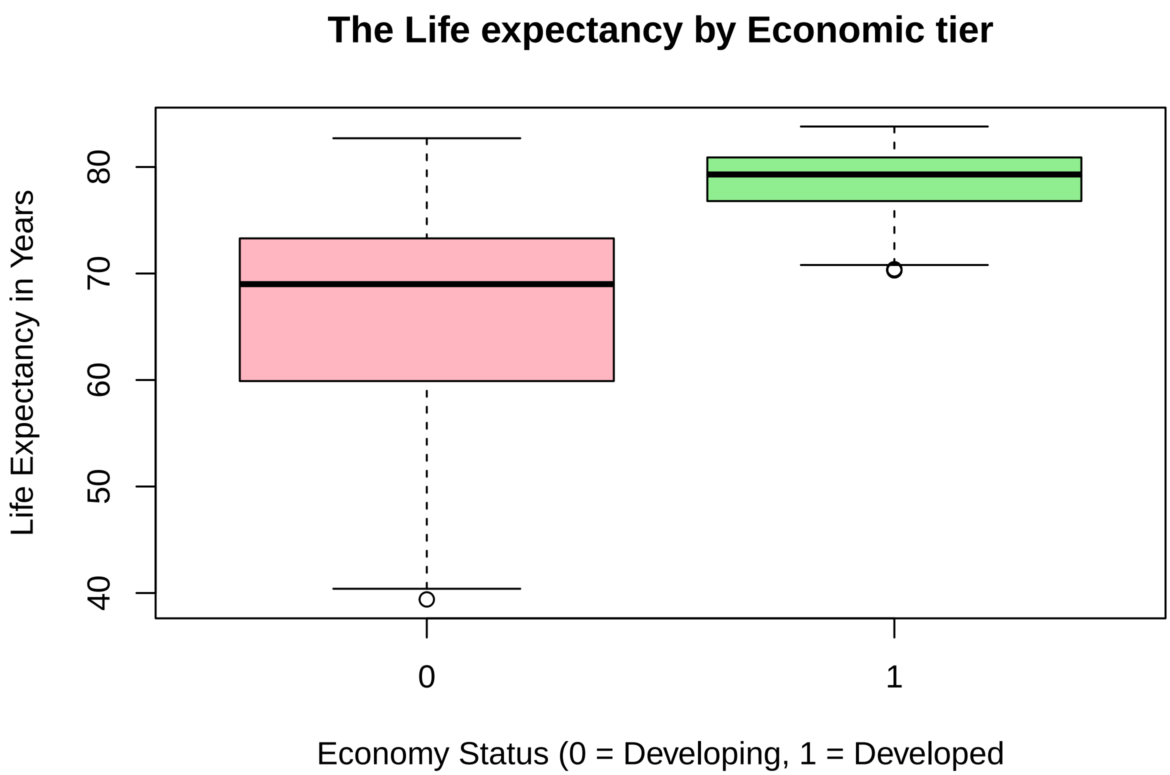

```r
# Incidents in HIV
hist(df$Incidents_HIV,
     main = "Distribution of HIV Incidents",
     xlab = "HIV incidents",
     col = "red",
     border = "white")
```



As previously explained by the numbers, the histograms revealed that the `GDP_per_capita`, `Population_mln` and `Incidents_HIV` were positively skewed.
The `Life_Expectancy` histogram was negatively skewed as expected, peaking at the ≈ 75 age mark.

We proceeded to construct boxplots.

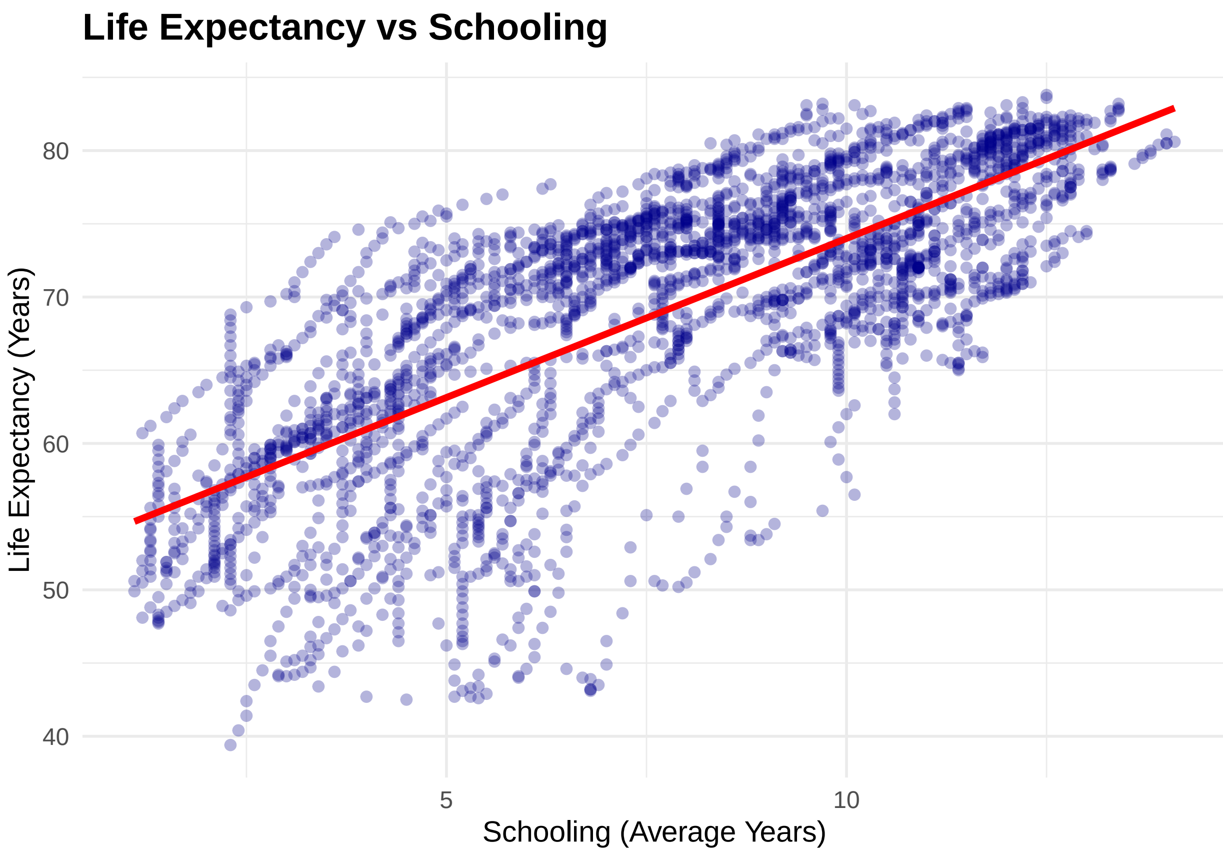
```r
# Life expectancy with economy status developed
boxplot(Life_expectancy ~ Economy_status_Developed,
        data = df,
        main = "The Life expectancy by Economic tier",
        xlab = "Economy Status (0 = Developing, 1 = Developed",
        ylab = "Life Expectancy in Years",
        col = c("lightpink", "lightgreen"))
```



From the boxplot, we saw that developed countries have a higher life expectancy than developing countries. This can as well be backed up by the vaccination coverage we identified previously. Developed countries have a higher coverage than developing countries.

We proceeded to construct a scatterplot of `Life_Expectancy` against `schooling`. We desired to visualize their relationship.

```
ggplot(df, aes(x = Schooling, y = Life_expectancy)) +
  geom_point(color = "darkblue", alpha = 0.3) +
  geom_smooth(method = "lm", color = "red", se = FALSE, linewidth = 1.2) +
  labs(title = "Life Expectancy vs Schooling", x = "Schooling (Average Years)", y = "Life
   ↪  Expectancy (Years)") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

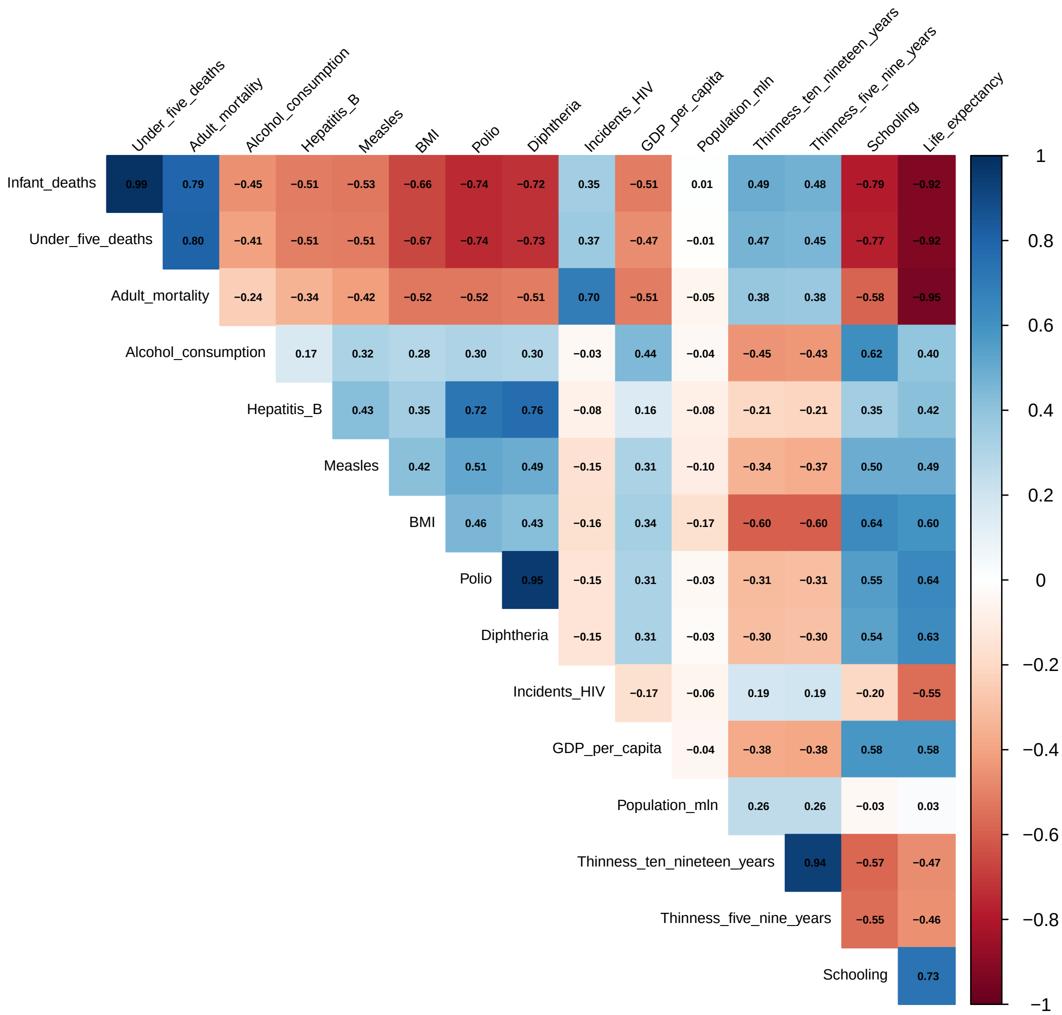


From the scatterplot, we saw that there was a positive relationship between `Life_expectancy` and `Schooling`. The more learned a country is, the higher the life expectancy. This is due to advanced education system yielding better research in how to make better health decisions as well as better standards of living.

We proceeded to have a correlation heatmap.

```
# For only numeric columns
cor_matrix <- cor(num_df, use = "complete.obs")

corrplot(cor_matrix,
         method = "color",
         type = "upper",
         addCoef.col = "black",
         number.cex = 0.5,
         tl.col = "black",
         tl.cex = 0.7,
         tl.srt = 45,
         diag = FALSE)
```



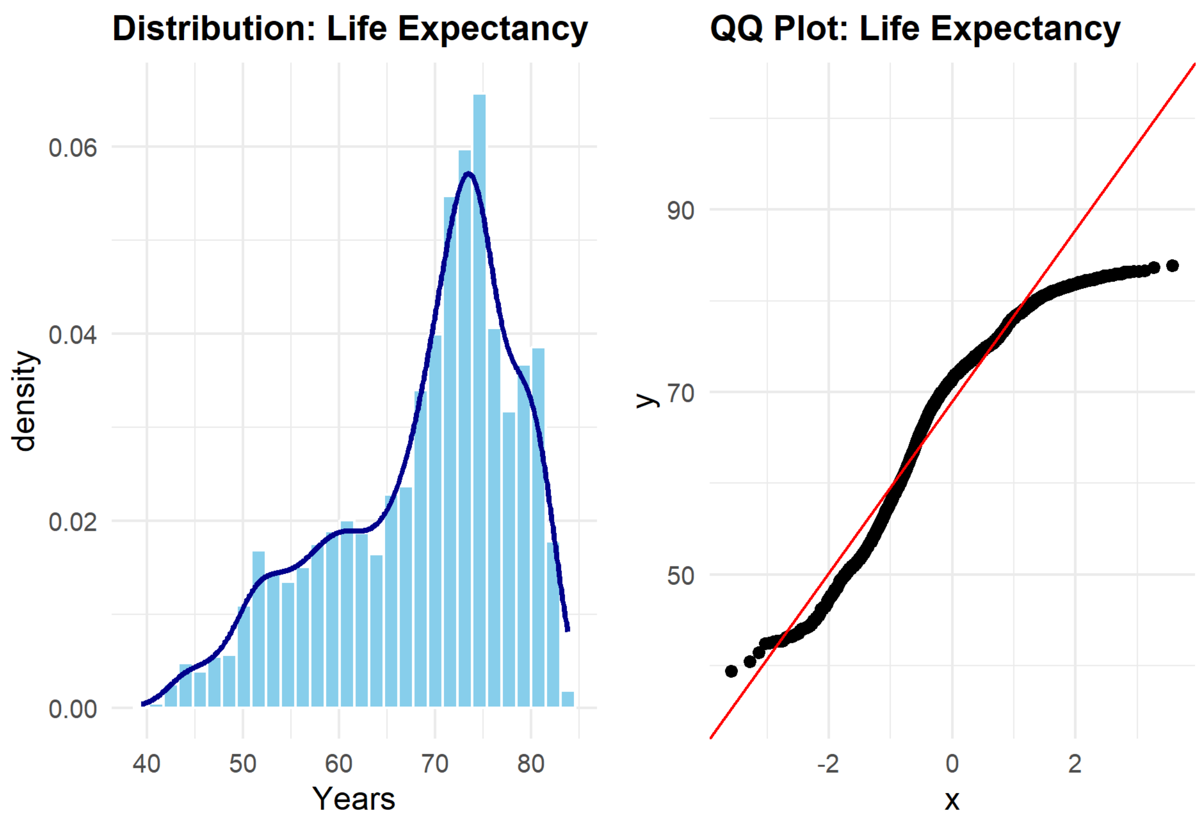
The heatmap validated the scatterplot as there there was a sufficiently high degree of correlation (`0.73`) between `Schooling` and `Life_expectancy`. We also saw a moderate positive relationship between `Life_expectancy` and `GDP_per_capita` as it was valued at `0.58`.
There was a very strong negative relationship (`0.95`) between `Life_expectancy` and `Adult_mortality`. This inverse relationship can be explained as the life expectancy increases, it results to a decrease in the rates of adult mortality.
We also saw evidence of multicollinearity amongst certain variables such as `Polio` and `Diphtheria`, `Infant_deaths` and `Under_five_deaths`, and `Thinness_ten_nineteen_years` and `Thinness_five_nine_years`. These variables almost had a perfect relationship. Subsequent steps would handle this to prevent our tests from crashing.

We proceeded to address the distributional assumptions. We selected 3 variables for this. We picked `Life_expectancy` as it was our dependent variable. Previously, we had observed that `GDP_per_capita` was skewed. This was also selected to see how it violated normality. We also selected a more symmetric variable which was `schooling` to serve as a comparison metric to the skewed variable.

```r
# Density plot + Histogram (Saved as p1)
p1 <- ggplot(df, aes(x = Life_expectancy)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, fill = "skyblue", color = "white") +
  geom_density(color = "darkblue", linewidth = 1) +
  labs(title = "Distribution: Life Expectancy", x = "Years")

# QQ Plot (Saved as p2)
p2 <- ggplot(df, aes(sample = Life_expectancy)) +
  stat_qq() + stat_qq_line(color = "red") +
  labs(title = "QQ Plot: Life Expectancy")

# Combine plots side-by-side using patchwork
p1 + p2
```




```r
# Shapiro-Wilk
shapiro.test(df$Life_expectancy)
```

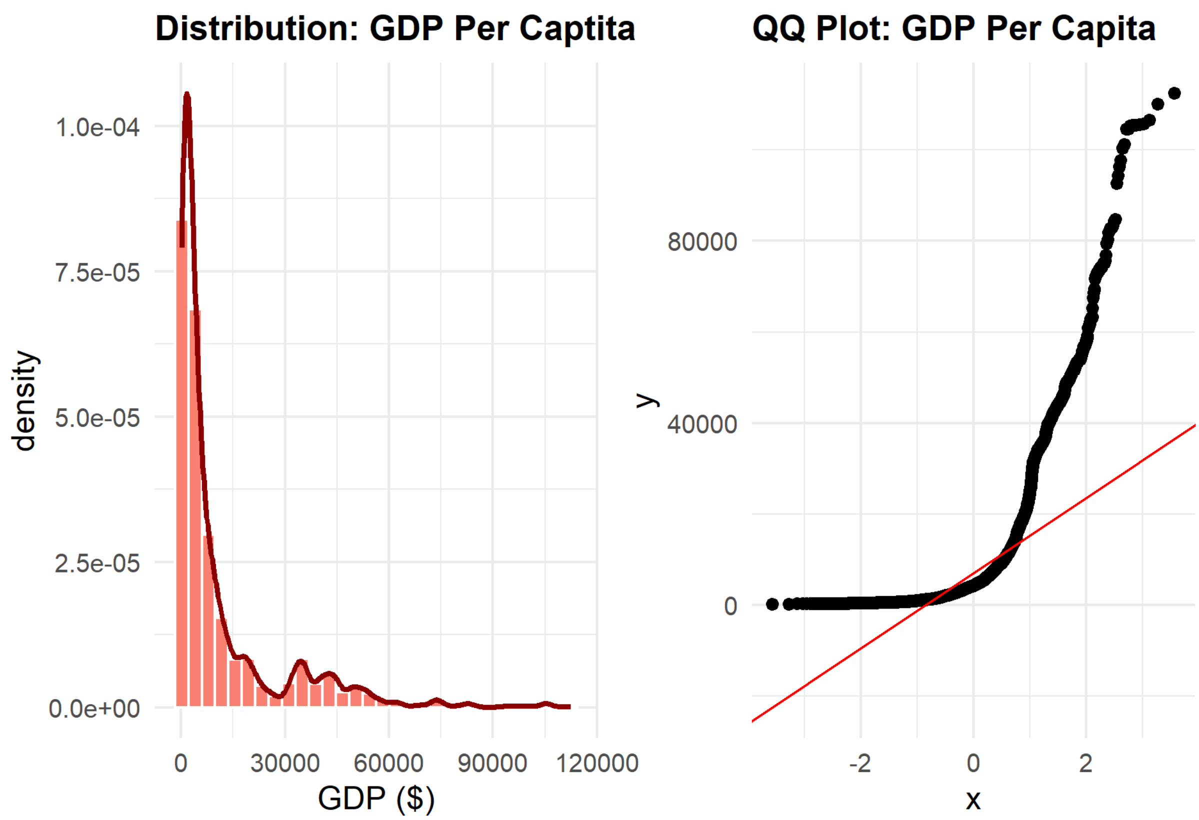
```
##
##  Shapiro-Wilk normality test
##
## data:  df$Life_expectancy
## W = 0.93497, p-value < 2.2e-16
```

The Histogram & Density plots showed that the life expectancy was negatively skewed with a peak at the ≈ 75 mark. The QQ Plot visualized the violation of normality as the plot had a heavy tail, deviating from the red line. Finally the shapiro-wilk test had a very low p-value thus indicating that normality is indeed violated. This would warrant a transformation before formally engaging in the various tests.
The next assessment was on the `GDP_per_capita`.

```r
# Density plot + Histogram
p1 <- ggplot(df, aes(x = GDP_per_capita)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, fill = "salmon", color = "white") +
  geom_density(color = "darkred", linewidth = 1) +
  labs(title = "Distribution: GDP Per Captita", x = "GDP ($)")

# QQ Plot
p2 <- ggplot(df, aes(sample = GDP_per_capita)) +
  stat_qq() + stat_qq_line(color = "red") +
  labs(title = "QQ Plot: GDP Per Capita")

# Combine plots
p1 + p2
```




```r
# Shapiro-Wilk
shapiro.test(df$GDP_per_capita)
```

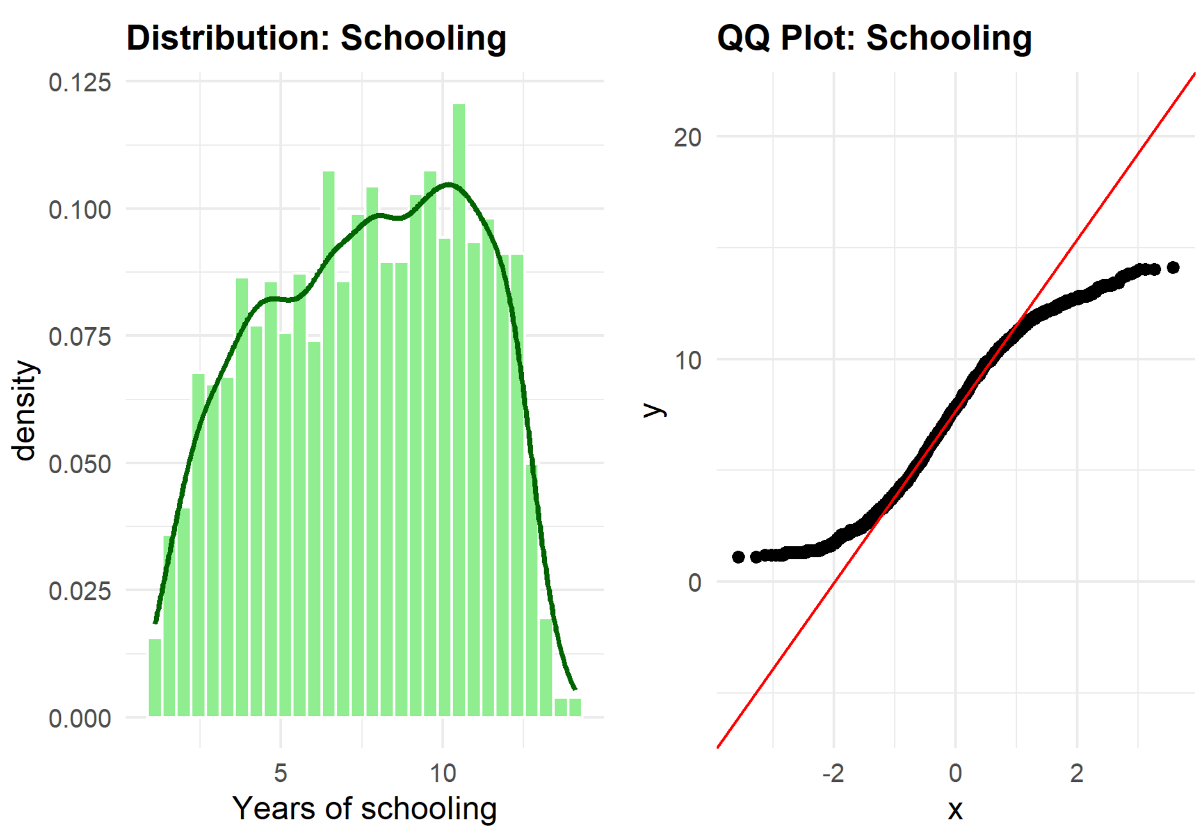
```
##
##  Shapiro-Wilk normality test
##
## data:  df$GDP_per_capita
## W = 0.67054, p-value < 2.2e-16
```

From the histogram and density plot, the `GDP_per_capita` was positively skewed. This was also evidenced by the QQ Plot which had much heavy tails as the points did not lie on the red line. The shapiro-wilk test gave a p-value close to zero indicative of the violation of normality.
We finally plotted for `schooling` variable.

```r
# Histogram and Density Plot
p1 <- ggplot(df, aes(x = Schooling)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, fill = "lightgreen", color ="white")
   ↪  +
  geom_density(color = "darkgreen", linewidth = 1) +
  labs(title = "Distribution: Schooling", x = "Years of schooling")

# QQ Plot
p2 <- ggplot(df, aes(sample = Schooling)) +
  stat_qq() + stat_qq_line(color = "red") +
  labs(title = "QQ Plot: Schooling")

# Combine plots
p1 + p2
```




```r
# Shapiro Wilk
shapiro.test(df$Schooling)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  df$Schooling
## W = 0.96809, p-value < 2.2e-16
```

From the Histogram & Density plot, the plot appeared to follow a uniform distribution. The QQ Plot showed the violation of normality as the plot had heavy tails and the Shapiro Wilk test had p-values close to zero, proving mathematically that normality was indeed in violation.

Before proceeding to formally execute the various tests onto our data, we had to address certain issues identified in the EDA phase. To begin with, from the heatmap, some of the variables were heavily correlated. In order to prevent our subsequent tests from collapsing and giving us not so useful outputs, we dropped them. Similarly, due to the severe skewness that was identified previously, some transformations were required. We proceeded to log transform `GDP_per_capita` and `Population_in_mln`. The former was necessary to capture the high range of differences between the lowest value and the highest value. The latter was also log transformed (as shown below) to counter the range (1379.78 - in millions) of population between highly and lowly populated countries. We also dropped the `Economy_status_Developing` variable and remained with `Economy_status_Developed` as we had determined previously that they were mutually exlusive.

```
# Dropping highly correlated features
df_clean <- df |>
  select(-Infant_deaths, # Kept Under_five_deaths
         -Diphtheria,     # Kept Polio
         -Thinness_five_nine_years, # Kept Thinness_ten_nineteen_years
         -Economy_status_Developing) # Kept Economy_status_Developed

# Log transformations
df_clean$Log_GDP <- log(df_clean$GDP_per_capita)
df_clean$Log_Population <- log(df_clean$Population_mln)

# Dropping the raw variables (original ones)
df_clean <- df_clean |>
  select(-GDP_per_capita, -Population_mln)

dim(df_clean)
```

```
## [1] 2864   17
```

4 variables were dropped as 2 variables were log transformed. This paved the way for the Parametric Tests.

Parametric Tests

1. Two-Sample T-test:Life Expectancy by Economic Status

Research Question: Is there a significant difference in the mean life expectancy between developed and developing nations?

Hypothesis
* **Null Hypothesis** $H_0 : \mu_{\text{developed}} = \mu_{\text{developing}}$ (There is no difference in mean life expectancy).
* **Alternate Hypothesis** $H_1 : \mu_{\text{developed}} \neq \mu_{\text{developing}}$ (There is a significant difference).

Our assumptions for this test were as follows:
* **Independence**: The observations are independent countries.
* **Normality**: Previously, it was determined that `Life_expectancy` did not follow a normal distribution but being guided by the Central Limit Theorem, it allowed us to continue with the Parametric test due to the large sample size ($N = 2864$)
* **Variance**: In order to account for the potential unequal variance of the two groups,we shall use the Welch's Two-Sample t-test which does not assume equal variance.

```
welch_test <- t.test(Life_expectancy ~ Economy_status_Developed, data = df_clean)

welch_test
```

```
##
```

```
##  Welch Two Sample t-test
##
## data:  Life_expectancy by Economy_status_Developed
## t = -53.685, df = 2623.8, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
## 95 percent confidence interval:
##  -12.60832 -11.71972
## sample estimates:
## mean in group 0 mean in group 1
##        66.34173        78.50574
```

The Welch two sample test was conducted to compare life expectancy between the developed and the developing nations. The project obtained the following results:
* `t = -53.685`: This is the test statistic. It measures how far apart the two group means are. This shows that there was a big difference between the means of the developing and developed nations. It was calculated as the mean life expectancy of developing nations minus the developed ones.
* `95 percent confidence interval: -12.60832 -11.71972`: This explains that with 95% confidence, the true population difference in life expectancy lies between developing and developed nations lies between 11.72 years and 12.61 years.
* `sample estimates: 66.34 against 78.51`: These are the sample means whereby the average life expectancy of developing nations is 66.34 years and that of developed nations is 78.51 years.

Since we obtained a p-value of less than 0.05, we reject the null hypothesis and conclude that developed nations have a significantly higher mean life expectancy than developing ones by approximately 12.17 years.

2. Analysis of Variance (ANOVA)

Research Question: Is there a significant difference in the mean life expectancy across different geographical regions in the world?

Hypothesis
* **Null Hypothesis** $H_0 : \mu_1 = \mu_2 = ... = \mu_k$ (There is no difference in mean life expectancy across different geographical regions).
* **Alternate Hypothesis** $H_1 : \mu_1 \neq \mu_2 \neq ... \neq \mu_k$ (There is a difference in mean life expectancy across different geographical regions or at least one geographical mean is different from the rest).

Our assumptions for this test were as follows:
* **Independence**: The observations are independent countries.
* **Normality**: Assumption of the residuals (difference between observed values and group means) are normally distributed as well as being guided by the Central Limit Theorem, it allowed us to continue with the Parametric test due to the large sample size ($N = 2864$)
* **Homogeneity of Variance**: ANOVA assumes that variance of each group is roughly equal (Homoskedasticity).

```r
anova_model <- aov(Life_expectancy ~ Region, data = df_clean)

summary(anova_model)
```

```
##                Df Sum Sq Mean Sq F value Pr(>F)
## Region          8 157346   19668   585.4 <2e-16 ***
## Residuals    2855  95930      34
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Post Hoc test - tells us which regions do differ that is
# if ANOVA is significant
```

```r
TukeyHSD(anova_model)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = Life_expectancy ~ Region, data = df_clean)
##
## $Region
##                                                   diff         lwr
## Asia-Africa                                11.60755719  10.53693803
## Central America and Caribbean-Africa       14.59019608  13.38115290
## European Union-Africa                      19.86774237  18.79712321
## Middle East-Africa                         16.12814251  14.77088500
## North America-Africa                       19.99436275  17.32193578
## Oceania-Africa                             11.67030971  10.17486937
## Rest of Europe-Africa                      16.67811275  15.35682949
## South America-Africa                       14.93342525  13.49014973
## Central America and Caribbean-Asia          2.98263889   1.63561619
## European Union-Asia                         8.26018519   7.03588534
## Middle East-Asia                            4.52058532   3.03908855
## North America-Asia                          8.38680556   5.64918787
## Oceania-Asia                                0.06275253  -1.54629271
## Rest of Europe-Asia                         5.07055556   3.62194444
## South America-Asia                          3.32586806   1.76518585
## European Union-Central America and Caribbean  5.27754630   3.93052360
## Middle East-Central America and Caribbean   1.53794643  -0.04647545
## North America-Central America and Caribbean  5.40416667   2.60950915
## Oceania-Central America and Caribbean      -2.91988636  -4.62417114
## Rest of Europe-Central America and Caribbean  2.08791667   0.53420041
## South America-Central America and Caribbean  0.34322917  -1.31547182
## Middle East-European Union                 -3.73959987  -5.22109664
## North America-European Union                0.12662037  -2.61099732
## Oceania-European Union                     -8.19743266  -9.80647790
## Rest of Europe-European Union              -3.18962963  -4.63824075
## South America-European Union               -4.93431713  -6.49499934
## North America-Middle East                   3.86622024   1.00431997
## Oceania-Middle East                        -4.45783279  -6.27027565
## Rest of Europe-Middle East                  0.54997024  -1.12167468
## South America-Middle East                  -1.19471726  -2.96436454
## Oceania-North America                      -8.32405303 -11.25401328
## Rest of Europe-North America               -3.31625000  -6.16126576
## South America-North America                -5.06093750  -7.96461955
## Rest of Europe-Oceania                      5.00780303   3.22214051
## South America-Oceania                       3.26311553   1.38539207
## South America-Rest of Europe               -1.74468750  -3.48689673
##                                                    upr     p adj
## Asia-Africa                                12.678176350 0.0000000
## Central America and Caribbean-Africa       15.799239259 0.0000000
## European Union-Africa                      20.938361535 0.0000000
## Middle East-Africa                         17.485400015 0.0000000
## North America-Africa                       22.666789708 0.0000000
## Oceania-Africa                             13.165750064 0.0000000
## Rest of Europe-Africa                      17.999395999 0.0000000
## South America-Africa                       16.376700760 0.0000000
## Central America and Caribbean-Asia          4.329661586 0.0000000
## European Union-Asia                         9.484485035 0.0000000
```

```
## Middle East-Asia                                6.002082087 0.0000000
## North America-Asia                             11.124423244 0.0000000
## Oceania-Asia                                    1.671797762 1.0000000
## Rest of Europe-Asia                             6.519166673 0.0000000
## South America-Asia                              4.886550261 0.0000000
## European Union-Central America and Caribbean  6.624568994 0.0000000
## Middle East-Central America and Caribbean     3.122368312 0.0651412
## North America-Central America and Caribbean   8.198824183 0.0000001
## Oceania-Central America and Caribbean        -1.215601587 0.0000040
## Rest of Europe-Central America and Caribbean  3.641632920 0.0010394
## South America-Central America and Caribbean   2.001930156 0.9993574
## Middle East-European Union                    -2.258103098 0.0000000
## North America-European Union                   2.864238059 1.0000000
## Oceania-European Union                        -6.588387423 0.0000000
## Rest of Europe-European Union                 -1.741018512 0.0000000
## South America-European Union                  -3.373634924 0.0000000
## North America-Middle East                      6.728120510 0.0009462
## Oceania-Middle East                           -2.645389930 0.0000000
## Rest of Europe-Middle East                     2.221615155 0.9839340
## South America-Middle East                      0.574930021 0.4764421
## Oceania-North America                         -5.394092781 0.0000000
## Rest of Europe-North America                  -0.471234243 0.0091737
## South America-North America                   -2.157255452 0.0000024
## Rest of Europe-Oceania                         6.793465547 0.0000000
## South America-Oceania                          5.140838992 0.0000027
## South America-Rest of Europe                  -0.002478271 0.0493486
```

The ANOVA test shows the mean life expectancy of at least one region is different from the other regions. On the other hand, the Tukey post-hoc test compares every region with all the other regions in order to obtain the gaps.

The anova obtained a p-value close to zero. Since the p-value was less than 0.05, we rejected the null hypothesis and concluded that there is a difference in mean life expectancy across different geographical regions.
The Tukey post-hoc test indicated that there were significantly large disparities between North America/EU/Asia and Africa. This showed that the former was outliving the latter by more than 10 years on average. For instance, `North America - Africa` had a difference of 19.99 years with a p-value very close zero indicating that it was significant. Non significant disparities were also identified such as `Oceania-Asia` which had a difference of 0.06 years with a p-value of 1.

3. Multiple Linear Regression

Research Question: By what value or extent do the economic indicators, healthcare of a country, physical metrics and demographic scales such as region predict the life expectancy of a country? Which of these variables are the most critical in determining the life expectancy of a country?

Hypothesis
* **Null Hypothesis** $H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$ (None of the predictor variables have a significant linear relationship with life expectancy).
* **Alternate Hypothesis** $H_1 : \beta_i \neq 0$ (At least one predictor is statistically significant).

Our assumptions for this test were as follows:
* **Linearity**: There is a linear relationship between dependent and independent variables.
* **Independence of errors**: Assumption of no autocorrelation (thus year will be omitted in the regression model).
* **Homogeneity of Variance**: Residuals must have a constant variance (Homoskedasticity).
* **Normality of Residuals**: Residuals must be normally distributed.

* **No multicollinearity**: The predictors are independent of each other.

```
# Removed country to prevent dummy variables as region
# takes care of country. Year omitted to prevent autocorrelation
df_reg <- df_clean |>
  select(-Country, -Year)

# Fitting the full model
full_model <- lm(Life_expectancy ~ ., data = df_reg)

summary(full_model)
```

```
##
## Call:
## lm(formula = Life_expectancy ~ ., data = df_reg)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.5670 -0.8148 -0.0138  0.7452  6.5284
##
## Coefficients:
##                                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        78.433382   0.655799 119.600  < 2e-16 ***
## RegionAsia                          0.025078   0.096381   0.260  0.79473
## RegionCentral America and Caribbean  1.928189   0.107083  18.006  < 2e-16 ***
## RegionEuropean Union               -0.674167   0.158882  -4.243 2.27e-05 ***
## RegionMiddle East                  -0.106298   0.123187  -0.863  0.38826
## RegionNorth America                 0.058726   0.215624   0.272  0.78537
## RegionOceania                      -0.600817   0.133436  -4.503 6.98e-06 ***
## RegionRest of Europe                0.443897   0.127186   3.490  0.00049 ***
## RegionSouth America                 1.302848   0.121439  10.728  < 2e-16 ***
## Under_five_deaths                  -0.075554   0.001509 -50.057  < 2e-16 ***
## Adult_mortality                    -0.046771   0.000559 -83.668  < 2e-16 ***
## Alcohol_consumption                -0.019039   0.010462  -1.820  0.06890 .
## Hepatitis_B                        -0.005217   0.002120  -2.461  0.01393 *
## Measles                             0.000621   0.001544   0.402  0.68750
## BMI                                -0.133424   0.020193  -6.607 4.66e-11 ***
## Polio                               0.008431   0.002949   2.859  0.00429 **
## Incidents_HIV                       0.085379   0.016668   5.122 3.22e-07 ***
## Thinness_ten_nineteen_years        -0.018458   0.007506  -2.459  0.01399 *
## Schooling                           0.088666   0.016704   5.308 1.19e-07 ***
## Economy_status_Developed            2.395997   0.142946  16.762  < 2e-16 ***
## Log_GDP                             0.489615   0.037253  13.143  < 2e-16 ***
## Log_Population                      0.155699   0.014817  10.508  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.195 on 2842 degrees of freedom
## Multiple R-squared:  0.984,  Adjusted R-squared:  0.9839
## F-statistic:  8309 on 21 and 2842 DF,  p-value: < 2.2e-16
```

From the `full_model`, we obtained a p-value of less than 0.05. We rejected the null hypothesis and concluded that at least one predictor is statistically significant. This was also evidenced by the numerous variables which had a p-value of less than 0.05.

However, there was a much bigger problem. The `Multiple R-squared` (0.984) and `Adjusted R-squared` (0.9839) were quite high. This showed that the model was explaining 98.39% of the variations. This was not a real world scenario as it indicated a high likelihood of overfitting and/or data leakage. In order to identify which variables had high multicollinearity, we proceeded to conduct the Variance Inflation Factor Test (VIF). Variables that had a VIF > 5 or > 10 would be candidates for omission.

```r
library(car)
cat("\n VIF Diagnostic \n")
```

```
##
##  VIF Diagnostic
```

```r
vif(full_model)
```

```
##                                  GVIF Df GVIF^(1/(2*Df))
## Region                      57.135455  8        1.287676
## Under_five_deaths            9.071305  1        3.011861
## Adult_mortality              8.270868  1        2.875912
## Alcohol_consumption          3.478939  1        1.865192
## Hepatitis_B                  2.305737  1        1.518465
## Measles                      1.663116  1        1.289618
## BMI                          3.934252  1        1.983495
## Polio                        3.965863  1        1.991447
## Incidents_HIV                3.158354  1        1.777176
## Thinness_ten_nineteen_years  2.224589  1        1.491506
## Schooling                    5.625837  1        2.371885
## Economy_status_Developed     6.718803  1        2.592065
## Log_GDP                      5.802112  1        2.408757
## Log_Population               1.603863  1        1.266437
```

The VIF test brought to life the inherent multicollinearity as explained below:
* `Under_five_deaths` and `Adult_mortality` had a 9.07 and 8.27 respectively. They are used to calculate life expectancy directly. This was a situation of data leakage as well as endogeneity. Both variables were dropped.
* `Economic_status_Developed` and `Log_GDP` both had values greater than 5. These variables essentially explain the same thing in that a country with a high GDP is in a higher economic tier. We dropped the categorical variable `Economic_status_Developed`. Hopefully this would lead to `Schooling` going down.
* `Region` was not omitted due its properties. The dataset had 9 different regions making the Degrees of Freedom to be 8. A series of dummy variables were created which in turned inflated the VIF to ≈ 57. The `GVIF^(1/(2*Df))` penalizes it due to the many dummy variables so that it could be observed on an equal mathematically playing field. Squaring it (1.287676) leads to ≈ 1.7, which is much below the 5 threshold.

```r
df_pruned <- df_reg |>
  select(-Under_five_deaths, -Adult_mortality, -Economy_status_Developed)

healthy_model <- lm(Life_expectancy ~., data = df_pruned)

summary(healthy_model)
```

```
##
## Call:
## lm(formula = Life_expectancy ~ ., data = df_pruned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.1971  -1.8175   0.0769   1.9846   9.8581
##
## Coefficients:
##                                         Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)                        21.639992   1.196965  18.079  < 2e-16 ***
## RegionAsia                          3.370484   0.238248  14.147  < 2e-16 ***
## RegionCentral America and Caribbean 5.173416   0.264031  19.594  < 2e-16 ***
## RegionEuropean Union                5.529740   0.314741  17.569  < 2e-16 ***
## RegionMiddle East                   2.228354   0.315028   7.074 1.89e-12 ***
## RegionNorth America                 3.262192   0.543812   5.999 2.24e-09 ***
## RegionOceania                       3.580553   0.324718  11.027  < 2e-16 ***
## RegionRest of Europe                5.349639   0.311107  17.195  < 2e-16 ***
## RegionSouth America                 4.668974   0.299024  15.614  < 2e-16 ***
## Alcohol_consumption                -0.291121   0.026279 -11.078  < 2e-16 ***
## Hepatitis_B                        -0.004631   0.005514  -0.840    0.401
## Measles                            -0.001750   0.004008  -0.437    0.662
## BMI                                 0.205423   0.049735   4.130 3.73e-05 ***
## Polio                               0.152926   0.006595  23.187  < 2e-16 ***
## Incidents_HIV                      -1.183328   0.029039 -40.749  < 2e-16 ***
## Thinness_ten_nineteen_years         0.038470   0.019356   1.987    0.047 *
## Schooling                           0.216259   0.041274   5.240 1.73e-07 ***
## Log_GDP                             3.128526   0.075681  41.338  < 2e-16 ***
## Log_Population                      0.325276   0.038000   8.560  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.109 on 2845 degrees of freedom
## Multiple R-squared:  0.8914, Adjusted R-squared:  0.8907
## F-statistic:  1298 on 18 and 2845 DF,  p-value: < 2.2e-16
```

```r
# Checking if VIF is below the 5 threshold
vif(healthy_model)
```

```
##                                  GVIF Df GVIF^(1/(2*Df))
## Region                      18.623803  8        1.200547
## Alcohol_consumption          3.242914  1        1.800809
## Hepatitis_B                  2.304150  1        1.517943
## Measles                      1.656692  1        1.287125
## BMI                          3.526199  1        1.877818
## Polio                        2.929921  1        1.711701
## Incidents_HIV                1.416373  1        1.190115
## Thinness_ten_nineteen_years  2.185736  1        1.478423
## Schooling                    5.075135  1        2.252806
## Log_GDP                      3.538162  1        1.881000
## Log_Population               1.558552  1        1.248420
```

From the model, we obtained a Multiple R-squared of 0.8914 and a adjusted R-squared of 0.8907. This informed us that the model explained 89.07% of the variation. This was a drop from the previous model which had been plagued by multicollinearity. VIF test was also conducted and we observed that schooling had a VIF of 5.07. As explained by **Grossman, 1972** and **Lleras-Muney, 2005**, dropping `Schooling` would introduce severe omitted variable bias. This is because it is a fundamental structural determinant of health. Additionally **Balaj, M., et al. (2024)** argues that every single additional year of schooling reduces the risk of adult mortality by 1.9%. Finally, a VIF of 5 falls well within acceptable limits according to standard econometric practices as explained by **O'Brien, 2007**. Its exclusion would misrepresent the independent causal effect that educational attainment exerts on life expectancy separate from income. We decided not to drop `Schooling` as it was essential for this project.

An interesting discovery was that upon the omission of the variables, the coefficients of `Incidents_HIV` became a negative (-1.18). Previously, it was a positive (≈ 0.08), indicating that the disease led to an increase in life expectancy and this was not the case.

We proceeded to construct the optimum model by using the stepwise function.This optimum model would aid in determining subsequent diagnostics to identify if other OLS assumptions were violated.

```r
optimum_model <- step(healthy_model, direction = "both", trace = 0)

summary(optimum_model)
```

```
##
## Call:
## lm(formula = Life_expectancy ~ Region + Alcohol_consumption +
##     BMI + Polio + Incidents_HIV + Thinness_ten_nineteen_years +
##     Schooling + Log_GDP + Log_Population, data = df_pruned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.2068  -1.8445   0.0865   2.0105   9.8184
##
## Coefficients:
##                                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        21.576062   1.194700  18.060  < 2e-16 ***
## RegionAsia                          3.387139   0.237611  14.255  < 2e-16 ***
## RegionCentral America and Caribbean  5.185287   0.263670  19.666  < 2e-16 ***
## RegionEuropean Union                5.556100   0.313119  17.744  < 2e-16 ***
## RegionMiddle East                   2.237065   0.314596   7.111 1.45e-12 ***
## RegionNorth America                 3.345999   0.533674   6.270 4.16e-10 ***
## RegionOceania                       3.621458   0.321483  11.265  < 2e-16 ***
## RegionRest of Europe                5.367197   0.308716  17.386  < 2e-16 ***
## RegionSouth America                 4.689453   0.297606  15.757  < 2e-16 ***
## Alcohol_consumption                -0.291137   0.026274 -11.081  < 2e-16 ***
## BMI                                 0.202428   0.049636   4.078 4.66e-05 ***
## Polio                               0.148623   0.004892  30.379  < 2e-16 ***
## Incidents_HIV                      -1.182048   0.028988 -40.777  < 2e-16 ***
## Thinness_ten_nineteen_years         0.038813   0.019329   2.008   0.0447 *
## Schooling                           0.215233   0.041189   5.225 1.86e-07 ***
## Log_GDP                             3.124673   0.075179  41.563  < 2e-16 ***
## Log_Population                      0.329366   0.037731   8.729  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.109 on 2847 degrees of freedom
## Multiple R-squared:  0.8914, Adjusted R-squared:  0.8908
## F-statistic:  1460 on 16 and 2847 DF,  p-value: < 2.2e-16
```

From the stepwise function, we obtained the combination of variables with the lowest AIC score. Some of the intercepts were interpreted as follows:
1. `RegionEuropeanUnion` had a coefficient of ≈ 5.55. This meant that holding all other factors constant, individuals living in the European Union have a life expectancy approximately 5.55 years higher than those living in the baseline region (Africa). This was the variable with the highest coefficient overall which meant that people living in the European Union have a higher life expectancy as compared to other regions.
2. `Alcohol_consumption` had a coefficient of ≈ -0.29. This meant that a unit change in the consumption of alcohol, would result in 0.29 reduction in life expectancy when all other factors are held constant. This explained that alcohol did not promote long life.
3. `Polio` had a coefficient of ≈ 0.14. This meant that a unit increase in Polio's vaccination coverage would result in an increase of 0.14 years in life expectancy when all other factors are held constant. Polio was the only vaccination disease that was picked by the optimum model which shows that its vaccination coverage is a great driver of life expectancy.

The optimum model had an adjusted R squared of 0.8908 which was a very slight increase from the regression model. This informed us that the model explained 89.08% of the variations as the remaining 10.92% would be explained by other factors not included in the model.

We proceeded to have a comparison table between the multiple linear regression model and the optimum model.

Table 1: Comparison of Healthy MLR and Optimum Stepwise Models

	Dependent variable:	
	Life_expectancy	
	Healthy Model	Optimum Model
	(1)	(2)
RegionAsia	3.370*** (0.238)	3.387*** (0.238)
RegionCentral America and Caribbean	5.173*** (0.264)	5.185*** (0.264)
RegionEuropean Union	5.530*** (0.315)	5.556*** (0.313)
RegionMiddle East	2.228*** (0.315)	2.237*** (0.315)
RegionNorth America	3.262*** (0.544)	3.346*** (0.534)
RegionOceania	3.581*** (0.325)	3.621*** (0.321)
RegionRest of Europe	5.350*** (0.311)	5.367*** (0.309)
RegionSouth America	4.669*** (0.299)	4.689*** (0.298)
Alcohol_consumption	−0.291*** (0.026)	−0.291*** (0.026)
Hepatitis_B	−0.005 (0.006)	
Measles	−0.002 (0.004)	
BMI	0.205*** (0.050)	0.202*** (0.050)
Polio	0.153*** (0.007)	0.149*** (0.005)
Incidents_HIV	−1.183*** (0.029)	−1.182*** (0.029)
Thinness_ten_nineteen_years	0.038** (0.019)	0.039** (0.019)
Schooling	0.216*** (0.041)	0.215*** (0.041)
Log_GDP	3.129*** (0.076)	3.125*** (0.075)
Log_Population	0.325*** (0.038)	0.329*** (0.038)
Constant	21.640*** (1.197)	21.576*** (1.195)
Observations	2,864	2,864
R^2	0.891	0.891
Adjusted R^2	0.891	0.891
Residual Std. Error	3.109 (df = 2845)	3.109 (df = 2847)
F Statistic	1,297.546*** (df = 18; 2845)	1,460.187*** (df = 16; 2847)
Note:		*p<0.1; **p<0.05; ***p<0.01

The adjusted R squared was still high for typical real world scenarios which indicated possible violations of the assumptions. Diagnostics were done onto these parametric tests to determine if indeed there were violations of the assumptions.